

NGUYEN PHUC THANH

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OBJECTIVE

As a hands-on full-stack multimedia engineer, I transform ideas into products—from production-ready Kotlin apps and polished web interfaces to scalable backends, smart ML models, and interactive Unity experiences. I get things done.

Check out my portfolio at: thanhng224.github.io/react-portfolio

EDUCATION

Hanoi University of Science and Technology – HUST

B.Sc. in Multimedia Engineering (Advanced Program)

Hanoi, Vietnam

Oct. 2021 – Aug. 2025 (Expected)

LICENSES AND CERTIFICATIONS

IELTS Academic - 7.0

Issued by IDP Education Co., Ltd

Sep. 2024

EXPERIENCE

ATIN – Advanced Technology Innovations

Full-Stack Developer (Mobile Focus)

Hanoi, Vietnam

Feb. 2025 – Present

- Developed and maintained Android face-terminal applications, covering end-to-end user flows for identity capture, biometric verification, and payment processing.
- Built a centralized device management solution with real-time monitoring, log aggregation, and automated update delivery to ensure reliable field operations.
- Prototyped cross-platform features to improve user engagement and streamline maintenance workflows.

FPT Software (Training Internship)

C++ Training Program

Hanoi, Vietnam

Sep. 2024 – Jan. 2025

- Completed an intensive C++ curriculum covering algorithms, data structures, software design, and system-level programming.
- Developed an AI-driven Tic-Tac-Toe game as a capstone project, enhancing problem-solving and teamwork skills.

PROJECTS

Below are just a few of my notable projects. Feel free to explore my GitHub for more!!!

Device Management System – Full-Stack Monitoring & OTA Update Platform | *Next.js, TypeScript,*

Node.js (Express), Socket.IO, MongoDB, Kotlin

Team Size: 1

GitHub: <https://github.com/ThanhNg224/DeviceManagement> (Private)

- Engineered a robust, end-to-end platform to manage and monitor Android-based face recognition terminals, presenting real-time telemetry (CPU, RAM, storage), firmware versioning, and connectivity status through a sleek dashboard.
- Architected a seamless OTA update pipeline: administrators upload signed APK packages via a Next.js interface, backend (Express + Socket.IO) broadcasts update events, and Kotlin agents on each device perform silent, background downloads and installations with Android's package manager.
- Designed a resilient heartbeat mechanism leveraging WebSockets for accurate online/offline detection, persisting detailed device logs and performance metrics in MongoDB to support historical analytics and proactive alerting.
- Developed an interactive log viewer featuring full-text search, multi-criteria filtering, pagination, and dynamic status badges, all secured with JWT-based authentication and granular role-based access control.
- Ensured scalability and maintainability through strict TypeScript typing, a modular codebase, and a comprehensive suite of unit and integration tests across both frontend and backend layers.

Viettel EID/Passport Verification App – Android Kiosk Solution | *Kotlin, Android, XML, ML Kit, REST API*

Team Size: 1

GitHub: <https://github.com/ThanhNg224/Viettel> (Private)

- Built a kiosk-based verification app on Joyusing Z10S Pro, orchestrating a 10-step CCCD/passport flow with document capture, OCR (MRZ) and NFC chip reading.
- Implemented portrait liveness checks and integrated ATIN Face Engine API for high-accuracy biometric matching.
- Developed digital signing and video-call verification modules, and integrated cash/QR-code payment flows with Viettel's BCCS backend.
- Structured app using MVVM with shared ViewModel, custom progress UI, and in-app feedback surveys to optimize UX.

Driving School App – Flutter Mobile App | *Flutter, Dart, Android Studio, Google MLKit (Face Detection)*

Team Size: 1

GitHub: <https://github.com/ThanhNg224/DrivingSchoolApp>

- Developed a Flutter-based educational app for driving schools, enabling video-based lessons with AI-driven face detection and automated engagement tracking.
- Implemented camera-based recording in 5-minute chunks, storing and compressing videos locally for future server upload and validation.
- Integrated Google MLKit to pause lessons when no face is detected, resuming upon user return for continuous engagement monitoring.
- Outlined a future server-side pipeline for automated identity verification and large-scale data management, ensuring the app's readiness for broader deployment.

Coffee Shop Website & Android App | *JavaScript, ReactJS, NodeJS, React Native, MySQL, Expo*

Team Size: 3

GitHub: <https://github.com/ThanhNg224/CoffeeShopProject>

- Built a coffee shop website and Android app with distinct user and admin roles, leveraging Node.js for the backend and MySQL for data.
- Implemented user features like product browsing, account management, rating/review, and feedback submission.
- Developed an admin panel for inventory management, sales data, user oversight, and review moderation.
- **My Role:**
 - * **Web:** Implemented login/logout with role management (admin vs. user), comments/ratings (publicly displayed), a site-wide view counter, and admin content moderation (edit pages, remove feedback).
 - * **Mobile App:** Connected the app to the backend via RESTful APIs, built main/login screens, and developed content display features that fetch data from the server.

Raincaster: Real-Time Forecasts with Advanced Rain Detection | *Python, TypeScript, Flask, ReactJS, Tailwind, scikit-learn*

Team Size: 1

GitHub: <https://github.com/ThanhNg224/Raincaster>

- Built a web-based weather forecasting platform integrating real-time meteorological data, with a dedicated focus on predicting rainfall.
- Architected a Flask-based backend that fetches and preprocesses real-time weather data for an SVM model.
- Developed a user-friendly interface displaying current conditions and predicted rain probabilities, helping users make timely decisions.

TECHNICAL SKILLS

Languages: JavaScript, Dart, Python, Kotlin, SQL

Mobile: Flutter, React Native, Android (Kotlin)

Front-end: ReactJS, NextJS, Tailwind CSS

Back-end: NodeJS, Flask, MySQL

Tools & Platforms: Firebase, Git, Linux, Blender, Maya

Machine Learning: scikit-learn, OpenCV

HONORS AND AWARDS

Third Prize in District Chemistry Competition

Issued by Haiphong Department of Education and Training

HaiPhong, Vietnam

Dec. 2020